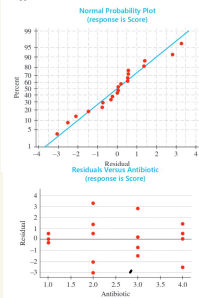


15.6

10. Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.² The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

Child	1	2	3	4
1	4.8	2.2	6.8	6.2
2	8.1	9.2	6.6	5.9
3	5.0	2.6	3.6	6.5
4	7.9	5.4	5.3	5.2
5	3.9	7.4	2.1	2.0

- a. What design is used in collecting these data?
b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the visual analysis of variance assumptions appear to be satisfied?



- c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

15.7

11. Rating Political Candidates A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance	x_i^2
1	12	22 (1)	144
2	7	165 (1)	49
3	5	300 (1)	25
4	19	15 (1)	361
5	17	180 (1)	289
6	12	25 (1)	144
7	9	14 (1)	81
8	16	60 (1)	256
9	3	230 (1)	9
10	8	200 (1)	64
11	15	130 (1)	225
12	4	13 (1)	16

- a. Calculate Spearman's rank correlation coefficient r_s .
b. Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

12. Competitive Running The number of years of competitive running experience related to a runner's distance running performance.³ The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table.⁴

Runner	Years of Competitive Running	10-Kilometer Finish Time (minutes)	x_i^2
1	9 (1)	33.15 (1)	81
2	13 (1)	33.33 (1)	169
3	5 (1)	33.50 (1)	25
4	7 (1)	33.55 (1)	49
5	12 (1)	33.73 (1)	144
6	6 (1)	33.80 (1)	36
7	4 (1)	33.85 (1)	16
8	5 (1)	34.15 (1)	25
9	3 (1)	34.50 (1)	9

- a. Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.
b. Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

a) This is a RBD (Randomized Block Design)

Treatment = 4 antibiotics

Blocks = 5 children

b) The normal probability plot looks fairly straight, but the residual plot shows different spread for different antibiotics

In the usual ANOVA assumptions are not clearly satisfied, especially equal variance.

c) H_0 : The response distributions are the same for 4 antibiotics
 H_A : At least one antibiotic has a different response distribution

Child	1	2	3	4
1	4.8 (1)	2.2 (1)	6.8 (1)	6.2 (1)
2	8.1 (1)	9.2 (1)	6.6 (1)	5.9 (1)
3	5.0 (1)	2.6 (1)	3.6 (1)	6.5 (1)
4	7.9 (1)	5.4 (1)	5.3 (1)	5.2 (1)
5	3.9 (1)	7.4 (1)	2.1 (1)	2.0 (1)

Rank within each child

Friedman statistic

$$F_r = \frac{12}{bk(k+1)} \sum R_j^2 - 3b(k+1)$$

$$F_r = \frac{12}{5(4)(5)} ((1^2+15^2+18^2+15^2) - 3(5)(5))$$

$$F_r = \frac{12}{100} (194+164+180+225) - 75$$

$$F_r = 8.12 (630) - 75$$

$$F_r = 76.56 - 75 = 1.56$$

$$X_{0.05,3}^2 = 7.815 \rightarrow 1.56 < 7.815 \text{ (Failed to reject } H_0)$$

→ There is not enough evidence to conclude that the taste response distributions differ among the four antibiotics.

11. Yield of Wheat Exercise 9 (Chapter 11

Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

Variety	1	2	3	4	5	6
A	35.3 (1)	31.0 (1)	32.7 (1)	36.0 (1)	37.2 (1)	33.1 (1)
B	30.7 (1)	32.0 (1)	31.4 (1)	31.7 (1)	35.0 (1)	32.7 (1)
C	38.0 (1)	33.1 (1)	33.4 (1)	37.1 (1)	37.1 (1)	36.2 (1)
D	34.9 (1)	36.1 (1)	35.2 (1)	38.3 (1)	40.1 (1)	36.0 (1)
E	32.4 (1)	28.9 (1)	29.2 (1)	30.7 (1)	33.6 (1)	32.1 (1)

- a. Use the Friedman F_r -test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.
b. When the analysis of variance was performed in Chapter 11, the ANOVA F -test produced the test statistic $F = 18.61$ with p -value = .000. How do these results compare with results of the Friedman F_r -test in part a? Explain.

a) H_0 : All five wheat varieties have the same yield distribution
 H_A : At least one variety has a different yield distribution

$$F_r = \frac{12}{bk(k+1)} \sum R_j^2 - 3b(k+1)$$

$$b = 6, k = 5$$

$$F_r = \frac{12}{6(5)(6)} ((1^2+11^2+24^2+21^2+7^2) - 3(6)(6))$$

$$F_r = \frac{12}{180} (1122) - 108 = 24.15$$

$$X_{0.05,4}^2 = 9.488 \rightarrow 24.15 > 9.488 \text{ (Reject } H_0)$$

→ There is sufficient evidence to conclude that the wheat varieties differ in yield

b) ANOVA

$$F = 18.61, p = 0.00 < 0.05 \text{ (Reject } H_0)$$

→ Both tests give same conclusion which reject H_0 , means that wheat variety has significant effect on yield

a) $H_0: \rho_s = 0$

$H_A: \rho_s < 0$

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

$$r_s = 1 - \frac{6(93)}{12(144-1)}$$

$$r_s = 1 - \frac{271.8}{1728}$$

$$r_s = 1 - 0.1574 \approx -0.1574$$

b) $\alpha = 0.05, n = 12$

$r_s \approx -0.497$

$$RR: r_s < -0.497$$

→ There is sufficient evidence to conclude that there is a negative rank correlation between rating & distance.

→ There is sufficient evidence to conclude that there is a negative rank correlation between rating & distance.

a) $H_0: \rho_s = 0$

$H_A: \rho_s \neq 0$

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

$$r_s = 1 - \frac{6(320.5)}{9(81-1)}$$

$$r_s = 1 - \frac{1231}{720}$$

$$r_s = 1 - 1.7095 \approx -0.7095$$

b) $\alpha = 0.05, n = 9$

$$|r_s|_{crit} = 0.683, |r_s| = 0.7095$$

$$RR: |r_s| > 0.683$$

→ There is sufficient evidence to conclude that there is a significant negative rank correlation between years of competitive running & 10-Km finish time.

→ There is sufficient evidence to conclude that there is a significant negative rank correlation between years of competitive running & 10-Km finish time.